CHEM 427/527 · SESSION 14 · TUESDAY, NOVEMBER 17, 2026

Uncertainty and model selection: which correction should the recess map use?

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Drafted with assistance by Claude Code. Reviewed and revised by Matt.

BY THE END OF THIS SESSION YOU CAN

Five things to do with a correction and its uncertainty

1. Choose between two models by their error on data held out of the fit. Hold the data out in groups, so that the test is as new as the data the model will meet.
2. Estimate a fitted correction’s uncertainty by a bootstrap over the unit that is independent. Say what the standard error from the covariance matrix leaves out.
3. Carry a correction’s uncertainty and the scatter of single days to a decision threshold. Count the cases the measurement cannot call.
4. Choose the direction of a calibration fit from the false alarms and misses each direction makes at the threshold.
5. Decide the resampling unit before asking the assistant for a bootstrap. Correct the first answer at once if it draws another unit, and record what it drew.

LAPTOPS AND DOWNLOADS, TO MINUTE 3

Run git pull, then start the download

In the terminal

```
git pull
```

The folder S14-model-selection appears, with notes.md, activity.ipynb, and data/ holding fetch_barkjohn.py, sensors.csv and README.md.

In activity.ipynb

Run the cells down to and including section 1’s first code cell, and leave it running. It downloads both EPA sets, 76 MB, and checks each file. With the room downloading at once, it takes about a minute.

A RuntimeError says what to do. If it persists, tell Matt.

Session 13’s notebooks are due at 1:30 pm today. Today’s are due at 1:30 pm on Thursday, November 19.

THE DECISION

Which correction should the recess map apply?

Who decides

A school district reads a map of low-cost PM2.5 sensors. A principal moves recess indoors when a sensor’s corrected 24 hour mean reaches 35.5 µg/m³, AQI 101, or 55.5 µg/m³, AQI 151, the red category. Someone chooses the correction the map applies.

A false alarm

Children lose recess on a day the air met the standard.

A miss

Children play outside in smoke on a day the reference put over the breakpoint.

Today you choose the correction for the district, from the counts at your own two sensors.

REAL DATA

Two EPA sets of PurpleAir sensors beside regulatory monitors

Dataset A, 24 hour means

12,413 rows

The rows come from 49 sensors in 15 states. Section 1 works the method on them, and the set is the same for everyone.

The smoke set, hourly

27 sensors

Many of them ran through the 2020 fire season. You have two of the 27, in section 2.

The raw reading at AQI 151

111 of 248

The uncorrected reading calls red on 248 of the smoke set’s 5,162 site-days. 111 of those calls are wrong, 52 of them on days that met the standard.

FROM SESSION 13

What did the standard error from the covariance matrix leave out?

The fit’s standard error

In session 13 the fitted crystallite size came with a standard error from pcov, the covariance matrix. It was a fraction of a percent of the size.

The certificate’s uncertainty

NIST’s certificate gave each reflection’s size an uncertainty of about a tenth of the size.

Which errors does a standard error from pcov not see?

THE ANSWER

What a standard error assumes

The standard error says how well these points pin this fit’s parameters, if every point is independent and the model is right. It says nothing about three things.

Points that err together

Points that share a source share its errors, so the data hold less information than their count suggests.

A different specimen

A new specimen, or a new sensor, was never in the fit.

One new measurement

The standard error does not say how far one new measurement scatters from the line.

Today: choose a correction by its error on held-out data, then give one corrected day its full uncertainty, σ.

LIVE: SECTION 1 ON DATASET A, ONE STATE LEFT OUT AT A TIME

Which error should choose the model?

Mean absolute error, µg/m³

Model	Held out	Own rows
Uncorrected	5.83	not fitted
Raw only	1.86	1.82
Raw + RH	1.59	1.57

The RH term

Relative humidity lowers the held-out error in 10 of the 15 states, so it earns its place in the correction.

A fit’s error on its own rows flatters it: the fit chose its coefficients to make exactly those errors small. Leaving out a whole state tests it the way a new city’s sensors will.

LIVE: THREE UNCERTAINTIES AT AQI 101, ON DATASET A

Which uncertainty sets σ?



All four bars are drawn to one scale, from 0 to 4.8 µg/m³.

At AQI 101

A raw reading of 76.1 µg/m³ at RH 50% corrects to 35.5 µg/m³.

The bootstrap over rows gives about twice pcov’s error: the scatter grows with concentration, and pcov assumes it does not.

One day’s scatter comes from the 390 rows corrected to 20 µg/m³ or more. It is nearly all of σ.

Rows are what pcov treats as independent. Is a row the unit that errs on its own?

ASSIGNMENTS

Two sensors each, by student number

#	Sensor 1	Sensor 2
1	Cedar Rapids	Marysville
2	Bend_S	Oakley_S
3	Appleton	Bend_S
4	Appleton	Tulelake_S
5	Missoula_S	Oroville_S
6	Atascadero	Wilmington
7	Marysville	Missoula_S
8	Oroville_S	Sarasota
9	Boise_S	Happy Camp_S
10	El Portal_S	Oakley_S
11	Keeler_S	Wilmington
12	El Portal_S	Topeka
13	Marysville	RTP_S
14	Atascadero	Decatur
15	Forks of S._S	Phoenix
16	Bend_S	Edmond
17	Boise_S	Decatur
18	Happy Camp_S	Oroville_S
19	Keeler_S	Oakridge_S
20	Edmond	Tulelake_S
21	RTP_S	Tulelake_S
22	Forks of S._S	RTP
23	Forks of S._S	Missoula
24	Boise_S	Phoenix
25	Cedar Rapids	Hoopa_S
26	Hoopa_S	Oakridge_S

Your two sensors are beside your student number from session 4 in data/sensors.csv. In Parameters, set **SENSORS** to the two names exactly as written, with the _S and the spaces, such as SENSORS = ["Keeler_S", "RTP"], and run the Parameters cell. Section 1 is the worked example on Dataset A, section 2 is your two sensors in the smoke set, and section 3 sets the inverted fit, EPA's published correction, beside the forward fit.

AI PRACTICE: CORRECT EARLY, AND CLEAR THE CONTEXT, FROM SESSION 8

Choose first, then read the answer

Say the choice

Before asking, say aloud the one choice the answer turns on. Here, a held-out test of this correction leaves out whole states.

The prompt, live

```
In my scratch notebook,
write a cross-validation of
raw_rh on train with five
folds.
```

Read for one thing

What does each fold leave out? If the answer splits rows at random, correct it at once, with the reason:

```
Leave out whole states, not
rows: deal the 15 states into
the five folds, with every row
of a state in one fold.
```

An answer that already leaves out states matches the choice: say what it drew, and move on.

AI PRACTICE: BEFORE TASK 3, THE BOOTSTRAP YOU WRITE

Decide the resampling unit, then ask

Decide

From how the data were collected, decide what each resample draws. Section 3 of the notes gives the rule.

Ask

Ask the assistant for the cell in your own words, and name the unit you chose.

Correct

If the first answer draws another unit, one sentence with your unit and its reason fixes it.

Clear

After two corrections that do not take, type /, choose Clear conversation, and start with a prompt that names the unit.

Your prompt record has three lines: your prompt, what the first answer drew and your correction, and what the given cell showed for your unit against rows. It counts toward the execution point.

THE JUDGMENT CELL, SECTION 4

Your fit for the district, and one entry

(a) Your answer

- As the analyst choosing the map’s correction: which fit should the district use?
- Cite, at your two sensors, the raw reading’s false alarms and your days within 2σ at both breakpoints, and, where the fits differ, each fit’s false alarms, with those on days that met the standard, and its misses.
- Say what the choice costs each way.

(b) One judgment entry

- Your prompt record is the source.
- Say what you did with a suggestion: changed, rejected or verified it.
- Cite a value from your output as the evidence.

Which can a principal less afford: several indoor days when the air met the standard, or one red day outside?

TO SUBMIT

Restart, Run All, save, upload by 1:30 pm on Thursday, November 19

Scored 0 to 3

Execution scores 1 when every cell runs after a restart and the notebook holds your prompt record. The judgment cell scores 0 to 2 by the rubric in the syllabus.

Submissions are individual

Work beside a neighbor for the pair Explain cell; each of you submits your own activity.ipynb to the Session 14 assignment on Canvas.

The third hour is an office hour. Chris grades session 14 and returns it by Wednesday, November 25.